Week 5

1-Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

```
#include<stdio.h>
int main(){
    int n;
    scanf("%d",&n);
    int a[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
    }
    for(int i=0;i<n;i++){
        for(int j=i+1;j<n;j++){
            if(a[i]==a[j]){
                printf("%d",a[i]);
                return 0;
            }
        }
    }
    return 0;
}
```

2-Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

```
#include <stdio.h>

int main() {
    int n, a[100], i;
    int slow, fast;

    scanf("%d", &n);

    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);

    slow = a[0];
    fast = a[0];

    do {
        slow = a[slow];
        fast = a[a[fast]];
    } while(slow != fast);

    slow = a[0];

    while(slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }

    printf("%d", slow);

    return 0;
}
```

*3-Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity*

```
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;
        scanf("%d", &n1);

        int a[n1];
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n2);

        int b[n2];
```

```
        for (int i = 0; i < n2; i++)
            scanf("%d", &b[i]);

        int i = 0, j = 0;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j])
                i++;
            else
                j++;
        }

        printf("\n");
    }

    return 0;
}
```

4-Print Intersection of 2 sorted arrays-O(m+n)Time Complexity,O(1) Space Complexity

```
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;
        scanf("%d", &n1);

        int a[n1];
        for (int i = 0; i < n1; i++)
            scanf("%d", &a[i]);

        scanf("%d", &n2);

        int b[n2];
        for (int i = 0; i < n2; i++)
            scanf("%d", &b[i]);

        int i = 0, j = 0;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                i++;
                j++;
            }
            else if (a[i] < b[j])
                i++;
            else
                j++;
        }

        printf("\n");
    }

    return 0;
}
```

5-Pair with Difference-O(n^2) Time Complexity,O(1) Space Complexity

```
#include <stdio.h>

int main() {
    int n, a[1000], k, i, j;

    scanf("%d", &n);

    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);
```

```
    scanf("%d", &k);

    for (i = 0; i < n; i++) {
        for (j = 0; j < n; j++) {
            if (i != j && a[j] - a[i] == k) {
                printf("1");
                return 0;
            }
        }
    }

    printf("0");

    return 0;
}
```

6-Pair with Difference-O(n) Time Complexity,O(1) Space Complexity

```
#include <stdio.h>

int main() {
    int n, a[1000], k;
    int i = 0, j = 1;

    scanf("%d", &n);

    for (int x = 0; x < n; x++)
        scanf("%d", &a[x]);

    scanf("%d", &k);

    while (j < n) {
        if (a[j] - a[i] == k && i != j) {
            printf("1");
            return 0;
        }
        else if (a[j] - a[i] < k) {
            j++;
        }
        else {
            i++;
        }

        if (i == j)
            j++;
    }

    printf("0");

    return 0;
}
```